SHARDA UNIVERSITY

School of Engineering and Technology | Department of Computer Science & Engineering

Course Code: CSCR2600 — Project-Based Learning (PBL) - 1 | Session: 2026–2027 (Sem 3, Sec B)

PROJECT DESIGN & EVALUATION MASTER DOCUMENT (EVALUATION-1)

Strictly Aligned with Rubric #R1 Criteria — Maximum Marks: 35 | Target Score: 35 / 35 (Grade A+)

Project Title:	**JAYMANI KRISHI SEWA** — Digital Agritech Marketplace & Bahi-Khata Ledger
Faculty Mentor:	**Prof. (Dr.) Ali Imam Abidi**, Professor, Dept. of Computer Science & Engineering
Project Team:	1. **Manish Pandey** (Roll No: 2515010348 | System ID: 2025266283) — Role: Lead Tester & Backend 2. **Jayant Raj** (Roll No: 2515010273 | System ID: 2025298708) — Role: Lead Designer & Full-Stack

1. OBJECTIVES AND METHODOLOGY OF PROJECT PROPOSAL (PO1) [MAX MARKS: 5]

1.1 Problem Formulation & Grassroots Realities:

- **Blind Physical Stock Hunting:** Rural farmers (District Gonda, UP) commute 15–30km across fragmented markets during sowing season, only to discover government-subsidized fertilizers (Urea, DAP) are out of stock.
- **Economic & Fuel Drain:** Each unconfirmed procurement trip wastes 4–6 productive farming hours and Rs. 150–350 in deadhead transportation fuel.
- **Middleman Price Opacity:** Due to lack of real-time price discovery, middlemen charge arbitrary markups above the Government MSP ceiling (e.g., selling subsidized Urea capped at Rs. 266.50 for Rs. 320–350).
- **Paper Bahi-Khata Ledger Disputes:** Over 80% of local agri-inputs rely on informal seasonal credit ('Udhaar'). Traditional paper notebooks suffer from physical weathering, alterations, calculation errors, and trust breakdown.

1.2 Core Engineering Objectives:

1. **Sub-100ms Hyper-Local Stock Visibility:** Deliver real-time catalog discovery and geodetic distance within 5km radius under 100ms.
2. **Tamper-Proof Digital Credit Ledger Engine:** Double-entry accounting enforcing mathematical balance invariance, credit ceiling checks (Rs. 10,000 pool), and SHA-256 audit hashes.
3. **Verified Price Transparency:** Continuous price cap monitoring cross-referenced with District Agriculture Officer (DAO) MSP caps and National SeedTrace Portal (seedtrace.gov.in).
4. **Click-to-Reserve & Central Escrow:** Atomic stock locking, dynamic UPI QR (jaymani.escrow@upi), and dynamic 4-digit OTP handover.

1.3 Phased Agile Engineering Methodology (5-Step Lifecycle):

- **Step 1 (Grassroots Analysis):** Field surveys across 35 farming households & 6 dealers; defined requirements for bilingual UI, high-contrast badges, and low-bandwidth resilience.
- **Step 2 (Architecture & Schema):** 3NF relational schema design; Haversine spherical math modeling.
- **Step 3 (Modular Core):** Python Flask REST API backend; React.js & Tailwind CSS frontend.
- **Step 4 (Integration):** Razorpay HMAC SHA-256 webhooks; Central Escrow pipeline; ReportLab invoice PDFs.
- **Step 5 (Testing & UAT):** Multi-threaded concurrency stress testing; zero ledger drift verification across 1,000 runs.

1.4 End-to-End Operational Pipeline Flowchart (ASCII):

```
+-------------------------------------------------------------------------------------+ | FARMER CLIENT
APPLICATION | | [Radius <= 5km] [Select Category: Fertilizer / Seed / Chemical] |
+-------------------------------------------------------------------------------------+ | v HTTPS GET
/api/products?radius=5km
+-------------------------------------------------------------------------------------+ | FLASK REST API
GATEWAY (Nginx + Gunicorn) | | [Cache Check] -----> (Hit < 5ms: Redis) | (Miss: Native Haversine d <= 5km) |
+-------------------------------------------------------------------------------------+ | v
+-------------------------------------------------------------------------------------+ | MYSQL DATABASE
(ACID Inventory & Spatial Query) |
+-------------------------------------------------------------------------------------+ |
+---------------------+---------------------+ | [Path A: Online Pay] | [Path B: Credit / Udhaar] v v
+-------------------------------------+ +-------------------------------------+ | RAZORPAY PAYMENT
GATEWAY | | DIGITAL BAHI-KHATA LEDGER CHECK | | - Render Escrow UPI QR (jaymani@upi) | | - Validate Debt <=
Rs.10,000 Credit Cap | | - HMAC SHA-256 Webhook Verification | | - Atomic Row-Level Balance Update |
+-------------------------------------+ +-------------------------------------+ | |
+---------------------+---------------------+ v
+-------------------------------------------------------------------------------------+ | ATOMIC ORDER &
CENTRAL ESCROW FULFILLMENT | | 1. Lock Row (SELECT ... FOR UPDATE) 2. Decrement Stock (S = S - Q) | | 3. Mint
Dynamic 4-Digit OTP (e.g. 1841) 4. Physical Handover -> Funds Released |
+-------------------------------------------------------------------------------------+
```

2. IDENTIFICATION OF NECESSARY TOOLS / FORMULAS (PO5) [MAX MARKS: 5]

Tier	Technology	Version	Technical Justification & Role
Frontend Client	React.js, Tailwind CSS, Vite	18.2 / 3.4 / 5.1	Declarative component tree, sub-second rendering, zero bundle bloat for 3G.
Application Tier	Python Flask, Gunicorn WSGI	3.0.2 / 21.2	Stateless microservices, granular thread lock control, low memory overhead.
Database & ORM	MySQL Server, SQLAlchemy ORM	8.0.36 / 2.0	ACID consistency, foreign key cascades, pessimistic row locks (for_update).
Cache Layer	Redis In-Memory Key-Value	7.2.4	Sub-5ms response for spatial indices and JWT session token blacklist.
Payment & Doc	Razorpay API, ReportLab PDF	v2 / 4.1.0	HMAC SHA-256 webhook signatures, vector-clean digital tax receipts.
DevOps	Docker, Nginx Reverse Proxy	24.0 / 1.24	Reproducible container execution, rate limiting, and SSL/TLS termination.

2.2 Mathematical Formulations & Engineering Equations:

```
1. Bahi-Khata Running Balance Equation: B_current = B_prev + Sum(P_j * Q_j) - R_repay Constraint: Approved iff
(B_current <= L_sanctioned), where L_sanctioned = Rs. 10,000.00 2. Thread-Safe Inventory Decrement Formula:
S_available = S_total - (S_reserved + S_confirmed) Update Rule: Delta(S_available) = -Q_order iff (S_available
>= Q_order) within atomic transaction. 3. Haversine Proximity Geofencing Formula: a = sin^2(Delta_lat / 2) +
cos(lat1) * cos(lat2) * sin^2(Delta_lon / 2) d = 2 * R * arcsin(min(1, sqrt(a))) [R = 6,371.0 km, Radius <= 5.0
km]
```

3. IDENTIFICATION OF THE MODULES (PO2) [MAX MARKS: 5]

Module Name	Primary Inputs	Core Processing Algorithms	Primary Outputs
Mod 1: RBAC Auth	Email, Password, Persona Role	PBKDF2 Password Verification, SeedTrace Portal Registry Validation	Signed JWT Token, Role Context, Certified Badge
Mod 2: Spatial Discovery	Latitude, Longitude, Radius (5km)	Haversine Spherical Filter, Redis In-Memory Spatial Grid Tiles	Sorted Store Array, Distance (km), Stock Volume
Mod 3: Vendor Inventory	SKU Code, Price, Stock Delta, Batch	District MSP Cap Compliance Check, ACID Atomic Stock Adjustment	Live Catalog JSON, Low-Stock Alerts (< 15 bags)
Mod 4: Bahi-Khata Ledger	Farmer ID, Udhaar / Repayment Sum	Double-Entry Math, Credit Ceiling Enforcement (<= Rs. 10,000)	Running Balance, SHA-256 Audit Hash, PDF Statement
Mod 5: Order & Escrow	Cart Items, Pay Method, 4-digit OTP	Pessimistic Row Lock, Dynamic OTP Minting, Escrow Release Logic	Order Ref, 5-Stage Stepper, Delivery Proof Receipt

4. WORKING WITH HIGH LEVEL DESIGN (HLD) (PO3) [MAX MARKS: 5]

4.1 3-Tier Enterprise Architectural Layout (ASCII):

```
+-------------------------------------------------------------------------------------+ | PRESENTATION TIER
(CLIENTS) | | [Farmer Mobile Client (React)] [Vendor Dashboard] [DAO Official Monitoring View] |
+-------------------------------------------------------------------------------------+ | HTTPS JSON Payloads
v +-------------------------------------------------------------------------------------+ | GATEWAY &
MIDDLEWARE TIER (NGINX + GUNICORN) | | - SSL/TLS Termination - Rate Limiting (Token Bucket) - CORS Filter |
+-------------------------------------------------------------------------------------+ | v
+-------------------------------------------------------------------------------------+ | APPLICATION TIER
(PYTHON FLASK ENGINE) | | +-----------------+ +-----------------+ +-----------------+ +----------------+ | | |
| Auth Guard (JWT) | | Spatial Service | | Inventory Engine | | Ledger Engine | | | +-----------------+
+-----------------+ +-----------------+ +-----------------+ +----------------+ | | +-----------------------------------------+
+-----------------------------------+ | | | Order Stepper Service (5 Stages) | | Central Escrow & OTP
Service | | | +-----------------------------------+ +-----------------------------------+ |
+-------------------------------------------------------------------------------------+ |
+---------------------+---------------------+ v v +-------------------------------------+
+---------------------------------------+ | PERSISTENCE TIER (DATA & CACHE) | | EXTERNAL INTEGRATION TIER | |
- MySQL 8.0 Relational DB (ACID) | | - Razorpay UPI Central Escrow Pool | | - Redis In-Memory Spatial Cache | |
- Govt SeedTrace Portal (seedtrace.gov) | +-------------------------------------+
+---------------------------------------+
```

4.2 System Workload & Query Resource Distribution:

```
SYSTEM QUERY & COMPUTATIONAL WORKLOAD DISTRIBUTION [40%] Local Inventory Catalog & Spatial Geolocation Queries
/ / / [25%] / \ [20%] Bahi-Khata / \ Order Processing & Ledger Log / \ Payment Webhooks \ / \ / \ / \ / \ / \ /
[15%] User Auth & JWT Session Management
```

5. WORKING WITH LOW LEVEL DESIGN (LLD) (PO4) [MAX MARKS: 5]

5.1 Database Relational Schema (ER Specifications):

• **users:** (id INT PK, full_name VARCHAR(100), email VARCHAR(120) UNIQUE, password_hash VARCHAR(255), role ENUM, phone_number, village, created_at)
• **vendors:** (id INT PK, user_id FK->users.id, shop_name, proprietor_name, license_number, seedtrace_id, latitude, longitude, market_area, credit_pool_limit, is_verified)
• **products:** (id INT PK, vendor_id FK->vendors.id, sku_code, name, category, price, gov_msp_cap, stock_quantity, batch_number, is_active, *CHECK price <= gov_msp_cap*)
• **orders:** (id INT PK, order_reference UNIQUE, farmer_id FK->users.id, vendor_id FK->vendors.id, total_amount, payment_method, payment_status, delivery_stage, delivery_otp, receipt_hash)
• **bahi_khata_ledger:** (id INT PK, farmer_id FK->users.id, vendor_id FK->vendors.id, order_id FK->orders.id, entry_type, debit_amount, credit_amount, running_balance, entry_hash, timestamp_ist)

5.2 REST API Endpoint Contracts:

```
CONTRACT 1: GET /api/products?radius=5km■=27.1352&lon;=82.0124 (200 OK) Response: {"success": true, "products":
[{"product_id": 101, "name": "Neem Coated Urea (45kg)", "price": 266.50, "gov_msp_cap": 266.50,
"stock_quantity": 85, "vendor": {"shop_name": "Pandey Traders", "seedtrace_id": "24-076000", "distance_km":
1.8}}]} CONTRACT 2: POST /api/ledger/credit-order (201 Created) Payload: {"farmer_id": 1, "vendor_id": 1,
"purchase_amount": 1340.00, "order_reference": "JMKS-ORD-01"} Response: {"success": true, "status": "APPROVED",
"ledger_entry": {"prior_balance": 3200.00, "debit_amount": 1340.00, "new_running_balance": 4540.00,
"credit_limit": 10000.00}} CONTRACT 3: POST /api/payment/razorpay-webhook (200 OK) Payload: {"event":
"payment.captured", "payload": {"payment": {"entity": {"amount": 134000, "status": "captured"}}}} Response:
{"success": true, "escrow_status": "ESCROW_HOLD", "delivery_otp": "7194", "message": "Funds Locked"}
```

5.3 Dual-Branch Checkout Execution Flow (ASCII):

```
FARMER CLIENT FLASK API GATEWAY DATABASE & PAYMENT | | | |-- 1. Submit Cart Checkout ------->| | | |-- 2. Lock
Row (FOR UPDATE) ----->| | |<- 3. Row Lock Confirmed ---------| | | | |=================== BRANCH A: INSTANT
ONLINE PAYMENT ==================| | |-- 4A. Create Razorpay Order ---->| |<- 5A. Render Dynamic UPI QR ------|
| |-- 6A. Authorize UPI Payment ----->| | | |<-- 7A. Webhook: Signature Valid -| | |-- 8A. Mark ESCROW_HOLD
--------->| | |-- 9A. Decrement Stock (S = S-Q) ->| |<- 10A. Issue Delivery OTP (7194) -| | | | |
|================== BRANCH B: DIGITAL BAHI-KHATA (UDHAAR) ============| | |-- 4B. Query Outstanding Debt --->|
| |<- 5B. Debt: Rs.3200 (Limit 10k) -| | [Condition: Order (Rs.1340) + Debt (Rs.3200) <= Rs.10,000 Cap] | | |--
6B. Debit Ledger (New: Rs.4540)| | |-- 7B. Decrement Stock ----------->| | |-- 8B. MINT SHA-256 Entry Hash -->|
|<- 9B. Issue Bahi-Khata Invoice ---| | | | |=================== FULFILLMENT & PHYSICAL HANDOVER
==================| |-- 11. Present OTP at Store ------>| | | |-- 12. Validate OTP ------------->| | |-- 13.
Release Escrow to Vendor ->| |<- 14. Order Completed Receipt ----| |
```

6. COMMUNICATION (PRESENTATION) (PO10) [MAX MARKS: 5]

6.1 Core Defense Viva Q&A; with Technical Explanations:

Q1: Concurrency Conflict Resolution When 1 Unit of Stock Remains?
Defense: The architecture utilizes pessimistic database row-level locking via SQLAlchemy's .with_for_update() directive (emitting SQL SELECT ... FOR UPDATE). When Thread A requests the record, an exclusive lock is granted at the storage engine. Thread B enters a blocked wait-state. Thread A decrements stock from 1 to 0 and commits, releasing the lock. Thread B immediately unblocks, reads stock as 0, fails the constraint check, and is cleanly issued an HTTP 409 'INSUFFICIENT_STOCK' response with zero negative stock drift.

Q2: Mobile Connectivity Drop During Online Payment?
Defense: Payment capture does not rely on client-side callbacks. We implement asynchronous server-to-server webhooks via Razorpay. When the banking switch processes the payment, Razorpay sends an asynchronous HTTPS POST directly to /api/payment/razorpay-webhook. The backend verifies the HMAC SHA-256 signature, transitions order status to ESCROW_HOLD, and issues the 4-digit OTP. When the farmer regains connectivity, querying /api/orders immediately displays the active escrow voucher and OTP.

Q3: Tamper-Proof Append-Only Ledger Security Against Fraud?
Defense: The bahi_khata_ledger is an append-only ledger; direct UPDATE or DELETE operations are prohibited at the ORM layer. Every ledger row computes an internal SHA-256 hash linking the prior record's hash, farmer ID, debit amount, and timestamp: H_k = SHA256(H_{k-1} || FarmerID || Delta || Timestamp). Any manual database tampering breaks the cryptographic hash continuity, causing the administrative telemetry audit engine to immediately flag the discrepancy.

7. PROJECT DESIGN REPORT & ROADMAP (PO12) [MAX MARKS: 5]

7.1 Verification & Test Evaluation Matrix:

TC ID	Test Scenario	Execution Steps & Test Inputs	Expected Outcome	Status
TC-01	Role-Based Access Control Guard	Attempt accessing vendor dashboard with Farmer JWT	HTTP 403 Forbidden; unauthorized access blocked	**PASS**
TC-02	Concurrent Stock Race Condition	2 simultaneous threads checkout last DAP bag	Thread 1 succeeds; Thread 2 receives HTTP 409	**PASS**
TC-03	Bahi-Khata Synchronous Ledger Debit	Credit order Rs. 1,340; current balance Rs. 3,200	Debt balance rises to exact Rs. 4,540	**PASS**
TC-04	Cryptographic Webhook HMAC Check	Send forged payment payload with fake signature	Signature mismatch rejected (HTTP 400 Bad)	**PASS**

7.2 Future Scope & Long-Term Engineering Extensions:

• **Vernacular Voice Ordering (NLP):** Integration of Hindi/Bhojpuri automatic speech recognition (Whisper / IndicWav2Vec) for illiterate farmers.
• **AI-Driven Soil & Crop Inputs:** Coupling Soil Health Card telemetry with local weather forecasts to autonomously prescribe optimal fertilizer schedules.
• **WhatsApp Business API Micro-Bot:** Automated dispatch of Bahi-Khata ledger summaries and order collection OTPs directly via WhatsApp.

Candidate Signatures:

1. Manish Pandey (Roll No: 2515010348 | System ID: 2025266283)
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2. Jayant Raj (Roll No: 2515010273 | System ID: 2025298708)
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